LAST NAME: _____

Practice Quiz 08B — MATH 2055 — Fall 2026

Practice for Quiz 08

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

For a positive integer m, write $\mathbf{Z}_m$ for the set of congruence classes modulo m. Recall that $[x]_m = [y]_m$ if and only if $x - y = mk$ for some integer k.

1. (a) Show that the proposed rule

$$h : \mathbf{Z}_6 \longrightarrow \mathbf{Z}_4, \qquad h([x]_6) = [x]_4$$

is not well defined. Exhibit two representatives of the same input class that give different output classes.

(b) Prove that changing the codomain gives a well-defined rule

$$g : \mathbf{Z}_6 \longrightarrow \mathbf{Z}_3, \qquad g([x]_6) = [x]_3.$$